Kai Li

Curriculum Vitae

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RESEARCH INTERESTS

Reinforcement learning; statistical machine learning; sequential decision-making; simulation and optimization; experimental design; reproducible computational research.

EDUCATION

Stony Brook University **Expected May 2027**
Ph.D. in Applied Mathematics and Statistics, Statistics Track *Stony Brook, NY*
Research areas: reinforcement learning, statistical machine learning, simulation, and sequential decision-making.

Stony Brook University **December 2022**
M.S. in Applied Mathematics and Statistics *Stony Brook, NY*
Completed the academic requirements for both the Statistics and Operations Research tracks.

Stony Brook University **December 2022**
Advanced Graduate Certificate in Operations Research *Stony Brook, NY*

Stony Brook University **May 2022**
Advanced Graduate Certificate in Data and Computational Science *Stony Brook, NY*

The Ohio State University **May 2020**
B.S. in Mathematics, Theoretical Track *Columbus, OH*
Minors: Computer Information Science, Database Track; Economics, Theory Concentration.

PUBLICATIONS AND MANUSCRIPTS

Peer-Reviewed Publications

1. **Li, K.** and Zhu, W. "Combinatorial Game Theory and Reinforcement Learning in Cumulative Tic-Tac-Toe via Evaluation Functions." *Stats*, 9(2), Article 28, 2026. doi:10.3390/stats9020028. *First author.*
2. Xu, R., **Li, K.**, Wang, H., Kementzidis, G., Zhu, W., and Deng, Y. "RL-QESA: Reinforcement-Learning Quasi-Equilibrium Simulated Annealing." *2nd AI for Math Workshop at ICML 2025*, 2025. OpenReview. *Co-first author.*

Manuscripts in Preparation

- "Reinforcement Learning with Expanded Action Spaces in Blackjack: A Controlled Six-Algorithm Comparison." Manuscript in preparation with Taejin Park, Lichun He, Jeffery Liu, and Wei Zhu; co-first author.
- "Enhancing Regional Sea Level Predictions: A Unified Structural Equation Modeling Approach." Manuscript in preparation with Jiayou Chao, Guanchao Tong, Andreas Lietzau, Wuyin Lin, Benjamin Hechtman, Minghua Zhang, Weihao Wang, Yunlong Pan, Merry Ma, Sen Zhao, and Wei Zhu; coauthor.

RESEARCH EXPERIENCE

Reinforcement Learning for Blackjack Strategy **September 2025–Present**
Collaborative research project, Stony Brook University *Stony Brook, NY*

Publication study: Reinforcement Learning with Expanded Action Spaces in Blackjack: A Controlled Six-Algorithm Comparison **March 2026–Present**

- Compare first-visit on-policy Monte Carlo, off-policy Monte Carlo with weighted importance sampling, SARSA, Q-learning, deep Q-networks, and proximal policy optimization.
- Study three action settings: Hit/Stand, Hit/Stand/Double, and Hit/Stand/Double/Split.
- Use a common Gymnasium-compatible framework with state-dependent legal-action masks, replicated training, held-out evaluation, learning-curve analysis, measures of policy agreement, and matched decision-level simulations.
- Preparing a manuscript as a co-first author with Taejin Park, Lichun He, Jeffery Liu, and Wei Zhu.

Pilot study: Reinforcement Learning for Blackjack Strategy **September–December 2025**

- Compared first-visit Monte Carlo control with Exploring Starts and proximal policy optimization.
- Developed an infinite-deck Gymnasium-compatible environment supporting Hit, Stand, Double, Split, payouts for natural blackjacks, and state-dependent action legality.
- Evaluated the trained agents over 20,000 held-out rounds and identified limitations involving action-space comparability, split-hand credit assignment, reward variance, and deck assumptions.
- The pilot study motivated the subsequent controlled six-algorithm comparison.

Cumulative Tic-Tac-Toe: Combinatorial Game Theory and Reinforcement Learning **July 2024–March 2026**

Department of Applied Mathematics and Statistics, Stony Brook University *Stony Brook, NY*

- Introduced cumulative tic-tac-toe, a 3×3 variant in which play continues until the board is filled and players score all completed three-in-a-row sequences.
- Used combinatorial game theory to characterize the game and establish that it produces a draw under optimal play.
- Formulated the game as a Markov decision process and developed a temporal-difference learning framework for empirical policy evaluation.
- Designed a triplet-coverage difference evaluation function that reduced average training episodes by approximately 23.1% relative to random initialization.

Graduate Research Assistant **May 2025–August 2025**

Institute for Advanced Computational Science, Stony Brook University *Stony Brook, NY*

- Conducted a literature-based technical investigation of reinforcement learning for climate model parameterization and multi-agent adaptation to sea-level rise.
- Compared physics-based, data-driven, and reinforcement learning approaches and reviewed eight model-free algorithms with attention to sample efficiency, stability, hyperparameter sensitivity, and reproducibility.
- Analyzed a published multi-agent model of adaptation to sea-level rise involving government, residents, and businesses; formalized its states, actions, observations, stochastic costs, objectives, and Bellman equations.
- Outlined a preliminary multi-agent advantage actor–critic architecture and training pseudocode as a direction for future simulation studies, and prepared two technical presentations synthesizing the work.

RL-QESA: Reinforcement-Learning Quasi-Equilibrium Simulated Annealing **December 2024–July 2025**

Collaborative research project, Stony Brook University *Stony Brook, NY*

- Contributed to the formulation and evaluation of temperature control for simulated annealing using reinforcement learning.
- Represented temperature control as a block-level Markov decision process in which cooling is permitted after a quasi-equilibrium condition is satisfied.
- Used a transformer-based policy trained with proximal policy optimization to make adaptive cooling decisions from energy and acceptance statistics.
- Established theoretical conditions retaining classical convergence guarantees for simulated annealing and evaluated the method on Rosenbrock and Lennard–Jones optimization problems.

Undergraduate Research Assistant **September 2019–April 2020**

Mathematical Biosciences Institute, The Ohio State University *Columbus, OH*

- Studied deterministic and stochastic susceptible–infected–recovered models for epidemic dynamics using compartmental models, differential equations, survival functions, and cumulative hazard functions.
- Performed computational analyses of susceptible, infected, and recovered population trajectories and investigated how awareness effects alter epidemic behavior.

Research Assistant **July 2019–August 2019**

School of Mathematics, Sun Yat-sen University *Guangzhou, China*

- Reviewed methodological literature and phenotype summary statistics from genome-wide association studies.
- Studied univariate and multivariate methods for estimating relationships among phenotypes and distinguishing homogeneous from heterogeneous genetic effects; revised analytical designs and calculations based on preliminary findings.

TEACHING EXPERIENCE

Graduate Instructor and Teaching Assistant **August 2022–Present**

Department of Applied Mathematics and Statistics, Stony Brook University *Stony Brook, NY*

- Taught five undergraduate sections serving 240 students: AMS 394 Statistical Laboratory (Fall 2022 and Fall 2023)

and AMS 412 Mathematical Statistics (Spring 2023, Spring 2024, and Spring 2025).
- Supported four graduate sections serving 80 students: AMS 586 Time Series (Fall 2024, Fall 2025, and Fall 2026) and AMS 580 Statistical Learning (Spring 2026).
- Developed and delivered instruction in statistical theory, inference, R-based statistical computing, machine learning, time series, and applied data analysis; prepared course materials, assignments, assessments, and individualized feedback.
- Received two Excellence in Student Teaching awards, an Outstanding Student Teacher award, and two Teachers Rated Excellent Educators by their Students recognitions.

Teaching Assistant **July 2018–August 2018**
Education First *Guangzhou, China*
- Supported classroom activities, student feedback, and coordination with instructors.

RESEARCH MENTORING

Graduate Student Research Mentor **July 2026–August 2026**
Simons Summer Research Program, Stony Brook University *Stony Brook, NY*
Project: Independent Spatial and Temporal Evidence for Temperature-Mediated Marine Range Shifts
- Supervised Jack Lu's summer research project examining how bottom-water temperature may contribute to changes in marine species distributions along the U.S. Northeast Continental Shelf.
- Provided technical guidance on study design, statistical modeling, robustness checks, interpretation, and appropriate presentation of limitations.
- Guided scientific writing, data visualization, poster organization, and preparation for the final research presentation. Poster authors: Jack Lu, Kai Li, and Wei Zhu.

Mathematics Peer Mentor **August 2018–May 2020**
Department of Mathematics, The Ohio State University *Columbus, OH*
- Mentored mathematics and actuarial science students, encouraged participation in collaborative mathematical activities, connected students with academic and campus resources, and supported students facing academic or personal barriers.

HONORS AND AWARDS

Teachers Rated Excellent Educators by their Students (TREES), two-time recipient 2023, 2024
College of Engineering and Applied Sciences, Stony Brook University; recognized for academic years 2022–2023 and 2023–2024.

Excellence in Student Teaching, two-time recipient 2023, 2025
Department of Applied Mathematics and Statistics, Stony Brook University.

Outstanding Student Teacher 2024
Department of Applied Mathematics and Statistics, Stony Brook University.

Outstanding Academic Excellence in Applied Mathematics and Statistics 2022
Department of Applied Mathematics and Statistics, Stony Brook University.

Dean's List, four terms 2018–2020
The Ohio State University: Autumn 2018, Spring 2019, Autumn 2019, and Spring 2020.

Tumbleson Fund Scholarship 2018
Department of Mathematics, The Ohio State University.

SELECTED TECHNICAL PROJECTS

Scalable Statistical Computing on SeaWulf **September 2026–Present**
- Completed a large-scale regression inference study using 500,000 observations, comparing model-based linear regression, a standard bootstrap with 5,000 resamples, and the Bag of Little Bootstraps with five 100,000-row bags and 1,000 weighted resamples per bag.
- Parallelized the resampling workflow on Stony Brook University's SeaWulf cluster using Slurm and GNU Parallel with five Python processes; restricted numerical libraries to one thread per process and used fixed random seeds for reproducibility.
- Obtained closely aligned coefficient estimates and identical inferential conclusions across the three methods. BLB confidence-interval widths averaged 88.4% of the corresponding model-based widths in this dataset.

Importance Sampling Techniques for Variance Reduction **October–December 2022**
Studied importance sampling estimators and simulation designs for reducing Monte Carlo variance.

Time Series Forecasting of Store Sales **February–April 2022**
Compared ARIMA, recurrent neural network, LSTM, and GRU models; prepared time series data in R and Keras and evaluated forecasts using root mean squared error.

Factors Affecting the Wages of Adult Civilians in the United States **February–May 2022**
Conducted an econometric study of wage determinants using model specification, estimation, diagnostics, and interpretation.

Understanding Flight Delays **August–December 2021**
Analyzed 2012–2019 airline data using ridge regression, multilayer perceptrons, and k-nearest neighbors; compared predictive performance statistically and examined seasonal and airline-specific patterns.

Fractional Factorial Design **October–November 2021**
Analyzed a one-sixty-fourth fraction of a 2^{10} factorial design, including estimability, aliasing, and design interpretation.

PMLi: Statistical Methods for Partially Matched Samples **March–May 2021**
Developed and documented an R package implementing methods for pooling p-values from mixtures of independent and matched observations, missing data, and exceptional cases requiring standard procedures.

Gene-by-Stress Interaction and Depression Risk **April–May 2021**
Applied multiple regression methods to study the relationship among genetic factors, stress, and depression risk.

Study on the Efficacy of Nosocomial Infection Control (SENIC) **October–December 2020**
Collaborated on regression-based analysis of hospital infection control data, including model building, diagnostics, and interpretation.

Applications of Mathematics in Econometrics **August–November 2020**
Prepared a mathematical exposition connecting optimization, matrix methods, and statistical estimation in econometric analysis.

Online Bookstore Information Management System **January–April 2020**
Collaborated on the conceptual design and relational implementation of a bookstore database and information management system.

Additional early computational work: Rossmann store sales data integration and modeling; Microsoft malware exploratory analysis; Google Ngrams time series analysis; quasi-likelihood estimation; numerical computation of Euler's number; Conway's Game of Life in MATLAB; and implementations of Kruskal's algorithm, binary trees, and triplet-sum hashing.

SELECTED RESEARCH AND TECHNICAL PRESENTATIONS

1. **Li, K.** "An Introduction to Reinforcement Learning." Technical presentation, Department of Applied Mathematics and Statistics, Stony Brook University, April 23, 2026.
2. **Li, K.** "Applications in Reinforcement Learning: Extreme Weather and Climate Change — Deep Reinforcement Learning Formulation." Technical presentation, Institute for Advanced Computational Science, Stony Brook University, June 17, 2025.
3. **Li, K.** "Applications in Reinforcement Learning: Extreme Weather and Climate Change — Preliminaries." Technical presentation, Institute for Advanced Computational Science, Stony Brook University, June 3, 2025.
4. **Li, K.** "Cumulative Tic-Tac-Toe with Combinatorial Game Theory and Reinforcement Learning." Research presentation, Stony Brook University, October 30, 2024.
5. **Li, K.** "Game Theory in Games: Tic-Tac-Toe Research Progress." Research presentation, Stony Brook University, June 5, 2024.
6. **Li, K.** "Approximate Solution Methods in Reinforcement Learning: Tic-Tac-Toe Research Progress." Research presentation, Stony Brook University, September 20, 2023.
7. **Li, K.** "Reinforcement Learning Algorithms and Their Performance in Tic-Tac-Toe." Technical presentation, Stony Brook University, April 17, 2023.
8. **Li, K.** "Tabular Solution Methods in Reinforcement Learning with Applications, Part I." Technical presentation, Stony Brook University, March 21, 2023.
9. **Li, K.** "OpenSpiel: A Framework for Reinforcement in Games." Research-paper presentation, Stony Brook University, February 20, 2023.
10. **Li, K.** "Using Randomization to Break the Curse of Dimensionality." Research-paper presentation in Dynamic Programming, Stony Brook University, December 1, 2022; revised research presentation to a separate audience, Stony Brook University, January 25, 2023.

11. Li, S., **Li, K.**, and Suh, J. H. "Time Series Forecasting of Store Sales: ARIMA, RNN, LSTM, and GRU Time Series Modeling." Technical presentation in Statistical Learning, Stony Brook University, April 25, 2022.
12. Matsibekker, R., **Li, K.**, Hugo, C. S., and Green, T. "Google Ngrams." Technical presentation in Time Series, Stony Brook University, December 6, 2021.
13. **Li, K.**, Hyland, B., Yabor, V., Gueli, C., and Yao, P. F. "Quasi-Likelihood Estimation." Technical presentation in Categorical Data Analysis, Stony Brook University, May 3, 2021.
14. **Additional early presentations: Li, K.** "Investing Lessons from and Variations of Tic-Tac-Toe" (May 1, 2023); "Research in Applied Mathematics and Statistics and How the Research Relates to Life" (May 4, 2021); "Story of Science in Applied Mathematics and Statistics" (September 19, 2020); and "Boeing 737 MAX Crashes: Software's Role" (November 14, 2019).

TECHNICAL SKILLS

Programming and tools: Python, R, SQL, Slurm, GNU Parallel, Git, GitHub, Jupyter, RStudio, Visual Studio Code, LaTeX.
Machine learning and reinforcement learning: scikit-learn, TensorFlow/Keras, Gymnasium, Stable-Baselines3; Monte Carlo control, temporal-difference learning, deep Q-networks, proximal policy optimization, Markov decision processes.
Statistical and computational methods: regression, experimental design, nonparametric methods, time series analysis, simulation, Monte Carlo methods, optimization, simulated annealing, statistical computing, bootstrap methods, parallel computing, and high-performance computing.

SELECTED GRADUATE COURSEWORK

Reinforcement Learning; Statistical Learning; Big Data Analysis; Data Science Fundamentals; Time Series; Regression Theory; Simulation and Modeling; Stochastic Models; Statistical Computing; Design of Experiments; Mathematical Statistics; Dynamic Programming; Linear Programming.

PROFESSIONAL TRAINING

Responsible Conduct of Research in Engineering **Issued July 2022**
CITI Program; Credential ID 46910512